

On Quantitative Stability for the Isoperimetric Inequality

Known Approaches and Possible Developments

Matteo Gori

Supervisors:

Prof. Alessio Figalli

Prof. Luigi Ambrosio

September 14, 2026

The Sharp Isoperimetric Inequality

In 1958, De Giorgi proved the sharp Euclidean isoperimetric inequality in full generality: among all finite perimeter sets $E \subset \mathbb{R}^n$ with fixed volume, the perimeter $P(E)$ is uniquely minimized by round balls.

More recently a new question gained considerable interest:

Quantitative Stability

If a set has an almost optimal perimeter ($P(E) \approx P(B)$), does it need to be close to a ball?

Quantitative Isoperimetric Inequality

To formalize this, we compare a finite perimeter set E to a reference ball B satisfying $|B(r)| = |E|$.

Fraenkel Asymmetry

$$\alpha(E) = \frac{1}{|E|} \inf_{x \in \mathbb{R}^n} |E \Delta B(x, r)|$$

Isoperimetric Deficit

$$\delta(E) = \frac{P(E) - P(B)}{P(B)}$$

We seek the optimal growth rate for $C(n)$ such that:

$$\alpha(E) \leq C(n) \sqrt{\delta(E)}$$

Lower Bound: $C(n) \gtrsim \sqrt{n}$

Surprisingly, the expected optimal growth can appear in two very different regimes:

- **Small Asymmetry:** The computation for ellipsoids

$$E_\varepsilon = \left\{ \frac{x_1^2}{(1+\varepsilon)^2} + (1+\varepsilon)^{\frac{2}{n-1}} \sum_{i=2}^n x_i^2 \leq 1 \right\}$$

was performed in the thesis.

- **Large Asymmetry:** Let

$$E = B\left(-x, 2^{-\frac{1}{n}}\right) \cup B\left(x, 2^{-\frac{1}{n}}\right)$$

be the disjoint union of two distant balls, then

$$\delta(E) = 2^{\frac{1}{n}} - 1 \simeq \frac{\log(2)}{n} \text{ and } \alpha(E) = \frac{1}{2}.$$

Fuglede's Approach (1989)

Fuglede's original strategy restricted the analysis to small perturbations of the ball.

Definition

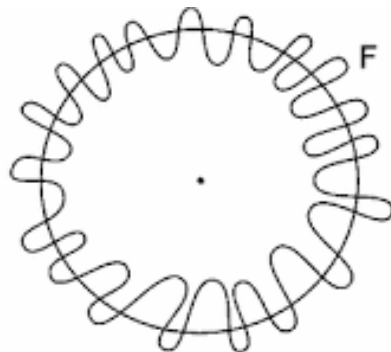
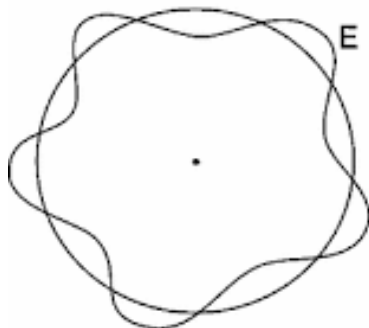
A set E is nearly spherical if there exists a Lipschitz function $u \in Lip(\mathbb{S}^{n-1})$ such that

$$E = \{s(1 + u(x))x : x \in \mathbb{S}^{n-1}, s \in [0, 1]\}$$

and moreover

$$|E| = \omega_n \quad \int_E x \, dx = 0$$

Nearly-spherical Sets



Fuglede's Result

Fuglede (1989)

There exist universal constants $C, \varepsilon > 0$ such that, if E is a nearly-spherical set with $\|u\|_{L^\infty} \leq \frac{\varepsilon}{n}$ and $\|\nabla u\|_{L^\infty} \leq \varepsilon$, then:

$$P(E) - P(B) \geq \frac{C}{\omega_n} |E \Delta B|^2.$$

- The proof relies on careful Taylor expansions of the perimeter functional.
- It exploits the interaction of u with the first spherical harmonics Y_0 and $Y_{1,j}$.
- It precisely gives $C(n) \lesssim \sqrt{n}$ in this regime.

The Idea of Fuglede's Proof

The area formula tells us that

$$P(E) = \int_{\mathbb{S}^{n-1}} (1+u)^{n-1} \sqrt{1 + \frac{|\nabla u|^2}{(1+u)^2}}$$

A suitable Taylor expansion, together with a linearization $\int u \simeq -\frac{n-1}{2} \int u^2$ of the volume condition gives

$$P(E) \geq \frac{1}{2} \int_{\mathbb{S}^{n-1}} |\nabla u|^2 - \frac{n-1}{2} \int_{\mathbb{S}^{n-1}} u^2 + h.o.t$$

The volume and barycenter conditions imply that $\int u \simeq 0$ and $\int u x \simeq 0$. Therefore, since the eigenvalues of the **spherical Laplacian** are $\lambda_k = k(k+n-2)$, we have approximately

$$2n \|u\|_{L^2}^2 \leq \|\nabla u\|_{L^2}^2$$

The First General Proof

Fusco-Maggi-Pratelli (2008) extended quantitative stability to all finite perimeter sets.

- A progressive sequence of symmetrizations reduces generic sets to more rigid geometries:

uniformly bounded $\rightarrow$ n -symmetric $\rightarrow$ axially symmetric

- The axially symmetric case is one dimensional in flavour, and in fact many different proofs were found, after the first one by Hall (1992).
- However, the *cut-and-paste* operations are rough and heavily rely on compactness arguments.

Approach via Optimal Transport

The most effective explicit estimate to date was achieved using mass transportation.

Figalli-Maggi-Pratelli (2010)

The constant is shown to grow as $C(n) \lesssim n^{\frac{9}{2}}$, which can actually be improved to n^3 .

- The proof employs a quantitative analysis of the **Brenier map** $T : E \rightarrow B$ from the set to a ball.
- It successfully proves stability even for the *anisotropic* isoperimetric inequality.
- It is widely believed this approach cannot be refined further to reach the expected $\sqrt{n}$ optimal growth.

The Selection Principle

Cicalese-Leonardi (2012) introduced a new proof strategy via regularity theory.

- ① A contradicting sequence E_j is replaced by minimizers F_j of the penalized functional:

$$\min \left\{ P(F) + \omega_n |\alpha(F) - \alpha(E_j)| + \Lambda ||F| - \omega_n| \right\} .$$

- ② The minimality of F_j implies they are **almost-minimizers**.
- ③ By Tamanini's theorem, these sets eventually gain $C^{1, \frac{1}{2}}$ regularity, becoming nearly-spherical and contradicting Fuglede's estimate.

Because it relies intrinsically on a contradiction argument, it provides **no explicit bound** for $C(n)$.

The Ideal Roadmap

To achieve the optimal growth $C(n) \lesssim \sqrt{n}$, we are following a "Dream Roadmap" suggested by Prof. Figalli.

Combine a **Refined Fuglede inequality** with a **Refined Selection Principle**.

More precisely:

- Extend Fuglede's result to require only small asymmetry, dropping the strict $W^{1,\infty}$ requirement.
- Find an improved Cicalese-Leonardi approach able to track the constants, or implement a well-behaved *continuous rounding process*.

The Sharp 2-Sobolev Inequality

This roadmap is heavily inspired by recent progress on the quantitative 2-Sobolev inequality, strenghtening the following classical result.

Aubin (1976), Talenti (1976)

For any $f \in \dot{H}^1(\mathbb{R}^n)$,

$$S_n \|f\|_{L^{2^*}(\mathbb{R}^n)}^2 \leq \|\nabla f\|_{L^2(\mathbb{R}^n)}^2 ,$$

where $2^* = \frac{2n}{n-2}$ and $S_n = \frac{1}{4}n(n-2)|\mathbb{S}^n|^{\frac{2}{n}}$ is the sharp constant.

Equality is attained exclusively by the **Aubin-Talenti bubbles**:

$$g_{a,b,y} = b \frac{1}{(1 + a|x - y|^2)^{\frac{n-2}{2}}}$$

The Quantitative Version

Dolbeault-Esteban-Figalli-Frank-Loss (2024) were able to bound the $\dot{H}^1$ -distance to the manifold of bubbles $\mathcal{M}$ using the deficit $\delta(f) = \|\nabla f\|_{L^2}^2 - S_n \|f\|_{L^{2^*}}^2$:

$$\inf_{g \in \mathcal{M}} \|\nabla f - \nabla g\|_{L^2(\mathbb{R}^n)}^2 \leq C(n) \left(\|\nabla f\|_{L^2(\mathbb{R}^n)}^2 - S_n \|f\|_{L^{2^*}(\mathbb{R}^n)}^2 \right)$$

To simplify the analysis, the problem is mapped to the sphere $\mathbb{S}^n$ via stereographic projection (adding a conformal factor).

Advantages of $\mathbb{S}^n$

In this framework:

- One of the Aubin-Talenti bubbles is mapped precisely to the **constant function 1**.
- By conformal invariance, we can assume that $d_{\dot{H}^1}(h, \mathcal{M})$ is achieved by 1, so that $(h = 1 + r)$

$$\|\nabla r\|_{L^2}^2 + A \|r\|_{L^2}^2 \leq C(n)(\|\nabla r\|_{L^2}^2 + A \|1 + r\|_{L^2}^2 - A \|1 + r\|_{L^{2^*}}^2)$$

where $A = \frac{n(n-2)}{4}$.

- By first variation, r additionally satisfies

$$\int_{\mathbb{S}^n} r \, d\sigma = 0 \qquad \int_{\mathbb{S}^n} r x \, d\sigma = 0$$

Idea of the Local Estimate

To control the norm $\|1 + r\|_{L^{2^*}}$, the function r is truncated into $r = r_1 + r_2 + r_3$ as follows:

$$r_1 = r \wedge \varepsilon_1 \quad r_2 = (r - \varepsilon_1)_+ \wedge (\varepsilon_2 - \varepsilon_1) \quad r_3 = (r - \varepsilon_2)_+ = r - r_2 - r_1$$

Intuitively:

- r_1 acts as a good approximation, hence it is controlled via the spectral gap (excluding degree 0 and 1 harmonics).
- r_2 behaves like a constant, hence satisfies an arbitrarily strong Poincaré inequality.
- r_3 absorbed and controlled by applying the sharp Sobolev inequality directly.

The Technical Tool

Lemma

There exists a constant $C_{\varepsilon_1, \varepsilon_2}$ such that for any $r \geq -1$ and any $2 \leq q \leq 3$ there exists a constant $C_{\varepsilon_1, \varepsilon_2}$ for which

$$\begin{aligned} & (1+r)^q \\ & \leq 1 + qr + \left(\frac{1}{2}q(q-1) + 2\varepsilon_1\theta \right) r_1^2 \\ & \quad + \left(\frac{1}{2}q(q-1) + C_{\varepsilon_1, \varepsilon_2}\theta \right) r_2^2 + \sum_{i \neq j} r_i r_j + (1 + \varepsilon_2\theta) r_3^q \end{aligned}$$

where $\theta = q - 2$.

It might look anonymous at first, but it is actually extremely sharp.

Consequence of the Lemma

After some simplifications, we can bound our deficit with

$$\frac{\varepsilon}{n}(\|\nabla r\|_{L^2}^2 + A\|r\|_{L^2}^2) + I_1 + I_2 + I_3$$

with

$$I_1 \simeq \|\nabla r_1\|_{L^2}^2 - n(1 + \varepsilon)\|r_1\|_{L^2}^2$$

$$I_2 \simeq \|\nabla r_2\|_{L^2}^2 - n(1 + C_\varepsilon)\|r_2\|_{L^2}^2$$

$$I_3 \simeq \|\nabla r_3\|_{L^2}^2 + A\|r_3\|_{L^2}^2 - A\|r_3\|_{L^q}^q.$$

and the approximation is up to terms carrying a factor $\theta = \frac{2}{n}$.

Finally our initial intuition shows that these terms are positive.

Local to Global

The "local-to-global" step in the Sobolev case is based on the following competing-symmetries result (where U is a specific conformal transformation).

Carlen, Loss (1990)

For any non-negative function $f \in \dot{H}^1(\mathbb{R}^n)$, the sequence $f_k = (\mathcal{R}U)^k f$ converges in $L^{2^*}(\mathbb{R}^n)$ to the Aubin-Talenti bubble

$$g = \frac{c}{(1 + |x|^2)^{\frac{n-2}{2}}}$$

with $c > 0$ such that $\|g\|_{L^{2^*}} = \|f\|_{L^{2^*}}$.

Continuous Symmetrizations

- The radial rearrangement can be partitioned into a sequence of **continuous Steiner symmetrizations**, a tool introduced by Brock (1995), defining a well-behaved flow.
- This flow converges to an Aubin-Talenti bubble, while satisfying **monotonicity** and **continuity** properties.

More precisely, the reduction works smoothly whenever we "cross the boundary" of the local case, i.e.

$$d_{\dot{H}^1}(f_t, \mathcal{M}) = \delta \|\nabla f_t\|_{L^2}^2$$

for some $t \in \mathbb{R}$ and a suitable (universal) $\delta > 0$.

Back to our Problem

To avoid relying purely on Taylor expansions, we shift the problem to $\mathbb{S}^n \subset \mathbb{R}^{n+1}$ and substitute $\mathbf{r} = (\mathbf{1} + \mathbf{u})^{\frac{n}{2}} - \mathbf{1}$.

$$\frac{P(E)}{(n+1)\omega_{n+1}} = \int_{\mathbb{S}^n} (1+r)^2 \sqrt{1 + \frac{4}{n^2} \frac{|\nabla r|^2}{(1+r)^2}} d\sigma$$
$$\frac{|E|}{\omega_{n+1}} = \int_{\mathbb{S}^n} (1+r)^q d\sigma \quad \text{where } q = 2 + \frac{2}{n}$$

- Mirroring the Sobolev case, we decompose $r = r_1 + r_2 + r_3$.
- We attempt to isolate small variations as a quadratic contribution, controlling them via spectral theory.

Some (weak) Good News

Hope

There exists a universal constant $\varepsilon > 0$ such that, if $\|r\|_{L^q} \leq \varepsilon$ and

$$\int_{\mathbb{S}^n} r \, d\sigma = 0 \quad \int_{\mathbb{S}^n} r x \, d\sigma(x) = 0,$$

then

$$C\theta \|r\|_{L^q}^2 \leq \delta(E)$$

for some universal constant $C > 0$.

Note that $\alpha(E) \lesssim \|r\|_{L^q}$, hence this would (almost) be a satisfactory local estimate.

In the thesis I motivated the validity of this result under the (very strict) additional hypothesis $\|\nabla u\|_{L^\infty} \leq \frac{\varepsilon}{\sqrt{n}}$.

Obstructions

We are first attempting to weaken the Lipschitz bound to $\|\nabla u\|_{L^\infty} \leq \varepsilon$. However, we encounter issues when dealing with r_3 .

Indeed, the computation quickly reduced to a Sobolev-like case by the expansion

$$(1+r)^2 \sqrt{1 + \frac{4}{n^2} \frac{|\nabla r|^2}{(1+r)^2}} \leq (1+r)^2 + \frac{2}{n^2} \left(1 + C \frac{\varepsilon}{n}\right) |\nabla r|^2$$

- The weaker bound introduces a perimeter loss factor proportional to $C\varepsilon$.
- However the $C\varepsilon$ contribution cannot be absorbed by terms with a factor of $\frac{2}{n}$, causing the functional decomposition to fail under these weaker hypotheses.

Algebraic Route

The failure of the previous attempt with $\|\nabla u\|_\infty \leq \varepsilon$ entirely comes from brutally expanding the perimeter.

We conclude that the only way to employ the truncation is to bound $P(E)$ only relying on monotonicities.

In this regard, central tools seems to be the monotonicity of

$$(1+r)^2 \sqrt{1 + \frac{4}{n^2} \frac{|\nabla r|^2}{(1+r)^2}} - (1+r)^2 = \frac{4}{n^2} \frac{|\nabla r|^2}{1 + \sqrt{1 + \frac{4}{n^2} \frac{|\nabla r|^2}{(1+r)^2}}}$$

and the sharp isoperimetric inequality stated as

$$\int_{\mathbb{S}^n} h^2 \sqrt{1 + \frac{4|\nabla r|^2}{n^2 h^2}} d\sigma \geq \|h\|_{L^q}^2$$

Obstacles in Local to Global

Analogously to the Sobolev case, there exists $s \in \mathbb{R}$ such that $\alpha(E_s) = \varepsilon$, hence $(\alpha(E) \leq 2)$

$$\frac{\alpha(E)}{\sqrt{\delta(E)}} \leq \frac{2}{\varepsilon} \frac{\alpha(E_s)}{\sqrt{\delta(E_s)}}$$

However, this scenario is drastically different, because we work with sets instead of functions.

More precisely, we are only restricting to sets which are round in measure, lacking any additional regularity properties.

Further Remarks

To reach Fuglede's graphicality conditions, a natural approach is to apply a (continuous) **star-shaped symmetrization**.

- Standard star-shaped symmetrization drastically decreases volume due to the t^{n-1} factor in spherical coordinates.
- Using a volume-fixing version is not viable: explicit counterexamples (Kawohl) show that the perimeter can actually *increase*.

Thank you for your attention!